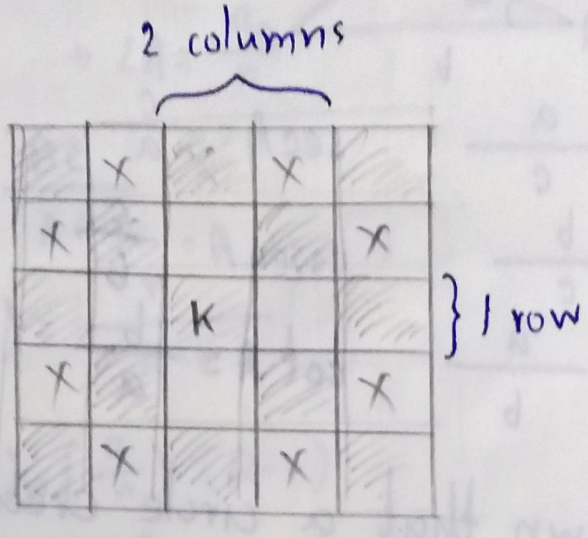
Problem 7: Given an $m \times n$ chessboard, find the max number of knights can be placed in a way such that no two knights attack each other. (Light OJ. 1010)

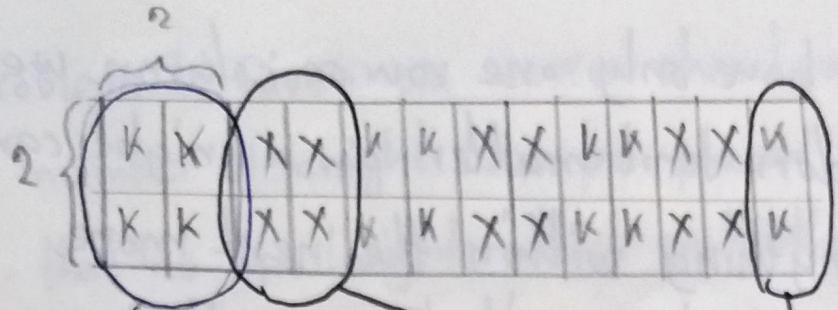
Solution:



If a knight is placed, the next move will occupy a block of opposite color. Also a knight can not take out a block of the same color. So, if the board is big enough, putting them on same color of block with most in number will be the amount of knights that can be placed. For the 5×5 board, $((5 \times 5) + 1)/2 = 13$ knights can be put as we will choose the block with most numbers.

However, if we have only one row or column, we can see that colors don't matter as a knight can not occupy anything within the next move, which row and column it is in. So the max number of knights that can be placed is the number of blocks in that single row/column.

Another corner case is if the number of rows or column is 2 and the other co-ordinating factor (column or row) is ~~greather~~ greater than 2 then knights can be filled up like 2X2 blocks occupying 2X2 blocks and having a 2X2 blocks of empty space from another occupying 2X2 block / 1X2 block depending on space availability.



1 occupying block 2X2

1 occupying block 1X2

1 empty block of 2X2

Pre-assumptions:

1. Familiarity with a chessboard and how a knight is permitted to move.